Wiyninger, Caleb
CSCI 3203: Tashfeen's Machine Learning I
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Homework 7

1) since $b^l_{new} = b^l_{old} - n(\frac{\partial error}{\partial b^l_{old}})$ then b^2 must be the same dimension in order for the subtraction to work. W follows the same rule, and the rule holds for all cases.
2) $\frac{df_n}{df_{n-1}} \cdot \frac{df_{n-1}}{df_{n-2}} \cdot \frac{df_{n-2}}{df_{n-3}} \cdots \frac{df}{dx}$
3)

$$W^{(l+1)^\dagger}(\frac{\partial error}{\partial b^{(l+1)}}) \circ \sigma`(z^l)$$
$$W^{(l+1)^\dagger}(W^{(l+2)^\dagger}(\frac{\partial error}{\partial b^{(l+2)}}) \circ \sigma`(z^{l+1})) \circ \sigma`(z^l)$$
$$W^{(l+1)^\dagger}(W^{(l+2)^\dagger}(W^{(l+3)^\dagger}(\frac{\partial error}{\partial b^{(l+3)}}) \circ \sigma`(z^{l+2})) \circ \sigma`(z^{l+1})) \circ \sigma`(z^l)$$
$$W^{(l+1)^\dagger}(W^{(l+2)^\dagger}(W^{(l+3)^\dagger} \cdots (W^{(l+n)^\dagger}(\frac{\partial error}{\partial b^{(l+n)}}) \circ \sigma`(z^{l+(n-1)})) \cdots \circ \sigma`(z^{l+2})) \circ \sigma`(z^{l+1})) \circ \sigma`(z^l)$$
$$W^{(l+1)^\dagger}(W^{(l+2)^\dagger}(W^{(l+3)^\dagger} \cdots (W^{L^\dagger}(\frac{\partial error}{\partial b^L}) \circ \sigma`(z^{L-1})) \cdots \circ \sigma`(z^{l+2})) \circ \sigma`(z^{l+1})) \circ \sigma`(z^l)$$
$$W^{(l+1)^\dagger}(W^{(l+2)^\dagger}(W^{(l+3)^\dagger} \cdots (W^{L^\dagger}(\frac{\partial error}{\partial a^L} \circ \frac{\partial a^L}{\partial z^L} \circ \frac{\partial z^L}{\partial b^L}) \circ \sigma`(z^{L-1})) \cdots \circ \sigma`(z^{l+2})) \circ \sigma`(z^{l+1})) \circ \sigma`(z^l)$$
$$W^{(l+1)^\dagger}(W^{(l+2)^\dagger}(W^{(l+3)^\dagger} \cdots (W^{L^\dagger}(2(a^L - t) \circ \sigma`(z^L) \circ 1) \circ \sigma`(z^{L-1})) \cdots \circ \sigma`(z^{l+2})) \circ \sigma`(z^{l+1})) \circ \sigma`(z^l)$$
$$W^{(l+1)^\dagger}(W^{(l+2)^\dagger}(W^{(l+3)^\dagger} \cdots (W^{L^\dagger}(2(a^L - t) \circ \sigma`(z^L)) \circ \sigma`(z^{L-1})) \cdots \circ \sigma`(z^{l+2})) \circ \sigma`(z^{l+1})) \circ \sigma`(z^l)$$

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Computer Science, Petree College of Arts & Sciences, Oklahoma City University